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SIR-models and uncertainty quantification

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Abstract

This thesis applies the theory of uncertainty quantification and sensitivity analysis on the SIR-model and SEIR-model for the spread of diseases. We attempt to determine if we can apply this theory to estimate the model parameters to an acceptable degree of accuracy.

Using sensitivity analysis we determine which parameters of the models are the most significant for some quantity of interest. We apply forward uncertainty quantification to determine how the uncertainty of the model parameters propagates to the quantities of interests. And lastly, we apply uncertainty quantification based on the maximum likelihood method to estimate the model parameters. To easily verify the results, we use synthetic data when estimating the parameters.

After applying these methods we see that the importance of the model parameters heavily depend on the choice of quantity of interest. We also note that the uncertainty method reduces the uncertainty in the quantities of interests, although there are a lot of sources of errors that still needs to be considered.

Sammanfattning

Denna kandidatavhandling tillämpar teorin bakom osäkerhetskvantifiering och sensitivitetsanalys på SIR-modellen och SEIR-modellen för spridning av epidemier. Vi utforskar om det går att applicera denna teori för att estimera modellernas parametrar till en acceptabel nivå.

Vi använder oss av känslighetsanalys för att avgöra vilka av modellens parametrar som har störst signifikans på en given kvantitet av intresse. Vi applicerar osäkerhetskvantifiering för att bestämma hur osäkerheten av modellens parametrar propagerar till kvantiteter av intresse. Till sist applicerar vi maximum likelihood-metoden för att estimera modellens parametrar. För att enkelt kunna verifiera resultaten används syntetisk data vid estimeringen.

Efter att metoderna har applicerats ser vi att betydelsen av modellens parametrar beror i stor utsträckning på valet av kvantitet av intresse. Vi kan även notera att osäkerhetsmetoderna reducerar osäkerheten för kvantitet av intresse, men medför några felkällor som man måste ta hänsyn till.

Acknowledgements

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1 Introduction

The spread of a pandemic depends on how people might behave and move, because people that are close to each other will infect each other. Trying to incorporate this random behaviour of people will lead to an increased complexity in creating a model. Instead, compartmental models were invented. In the models that will be discussed in this thesis, the behaviour and movement of people are idealized to different compartments. The effect of people meeting are modelled with different transfer rates between said compartments. One of the most prominent compartmental models is the SIR-model and its variations, which we will study in this thesis.

This thesis aims to explore uncertainty quantification of the SIR-model and the SEIR-model. We investigate whether it is possible to apply theory of uncertainty quantification to estimate the model parameters of the models, to a acceptable degree of accuracy. The thesis is limited to use synthetic data instead of real world data in order to make it possible to easily verify the results.

The thesis is structured in the following way. First we discuss some theoretical background for these compartmental models and how they can be approximated. Following that, we consider uncertainty in the parameters of the model and approaches for handling the challenges this introduces. Lastly, we concretely apply the mentioned methods on some numerical examples and discuss the results.

2 Theoretical background

2.1 The SIR-Model and the SEIR-model

The SIR-model is the simplest and most well known compartmental model. It depends on two parameters β and r and divides the population with N_{pop} individuals into three compartments: S (susceptible), I (infected), R (removed), $S + I + R = N_{pop}$. Parameter $\dfrac{\beta}{N_{pop}} \in [0, 1]$ is the transmission rate at which you go from susceptible to infected, and $r \in [0, 1]$ is the percentage transmission rate where you go from infected to removed. The proportion of the population in each category is described by the following system of ODE:s [2]

$$\begin{cases} \frac{dS(t)}{dt} &= -\frac{\beta}{N_{pop}} I(t)S(t), \\ \frac{dI(t)}{dt} &= \frac{\beta}{N_{pop}} I(t)S(t) - rI(t), \\ \frac{dR(t)}{dt} &= rI(t). \end{cases} \tag{1}$$

The SIR-model is a simple model which might not accurately describe all the details of a pandemic disease. In this thesis we also consider a variation of the SIR-model known as the SEIR-model. Another compartment is added: E (exposed). This compartment attempts to describe the incubation time, where one is infected, but is not yet infectious. We introduce a new parameter $i \in [0, 1]$ that is the transmission rate from exposed to infected, and we change β to be between susceptible and exposed. Resulting in the following system of differential equations [2]

$$\begin{cases} \frac{dS(t)}{dt} &= -\frac{\beta}{N_{pop}} I(t)S(t), \\ \frac{dE(t)}{dt} &= \frac{\beta}{N_{pop}} I(t)S(t) - iE(t), \\ \frac{dI(t)}{dt} &= iE(t) - rI(t), \\ \frac{dR(t)}{dt} &= rI(t). \end{cases} \tag{2}$$

Generally, these models are not solvable analytically and therefore one must apply numerical methods. We have decided to use the Euler forward scheme. Let $t_n = n\Delta t, n = 0, , 1, 2, \ldots$ with $\Delta t > 0$ and approximate $S(t_n) \simeq S_n, I(t_n) \simeq$

$I_n, R(t_n) \simeq R_n$ where S_n, I_n and R_n are defined as

$$\begin{cases} S_{n+1} &= S_n - \Delta t \left(\frac{\beta}{N_{pop}} I_n S_n \right), \\ I_{n+1} &= I_n + \Delta t \left(\frac{\beta}{N_{pop}} I_n S_n - r I_n \right), \\ R_{n+1} &= R_n + \Delta t \left(r I_n \right). \end{cases} \tag{3}$$

For the SEIR-model we similarly have the following Euler scheme

$$\begin{cases} S_{n+1} &= S_n - \Delta t \left(\frac{\beta}{N_{pop}} I_n S_n \right), \\ E_{n+1} &= E_n + \Delta t \left(\frac{\beta}{N_{pop}} I_n S_n - i E_n \right), \\ I_{n+1} &= I_n + \Delta t \left(i E_n - r I_n \right), \\ R_{n+1} &= R_n + \Delta t \left(r I_n \right). \end{cases} \tag{4}$$

Given some initial conditions for these variables, we obtain implementations with approximations for the SIR-model (1) and the SEIR-model (2) using (3) and (4).

2.2 Forward Uncertainty Quantification

There is a difficulty in knowing the exact values for the model parameters (β, r, i), which creates a need to estimate them and will add additional errors to the models outcome. The uncertainty in the parameters allow us to consider these parameters as random variables under certain distributions. To make forecasts we want to know how much the randomness of these variables affect the quantities of interest of our models. Where a quantity of interest is the quantity we want to determine. An example is the maximum amount of infected people or at what time the infected peak happened. This can be done using theory of uncertainty quantification. There are a multitude of alternative methods such as calculating the sample mean and variance of the quantities of interest, or estimating the full probability density function.

2.2.1 Kernel Density Estimation

A Kernel Density estimation can be used to construct a probability density function (pdf) for a set of N independent samples $\{x_i \in \mathbb{R}, i = 1, \ldots, N\}$ with an unknown distribution. The pdf estimation is given by the formula

$$\hat{f}_h(x) = \frac{1}{Nh} \sum_{n=1}^{N} K\left(\frac{x - x_i}{h}\right), \tag{5}$$

as stated in [1]. The function K is called the kernel function which is a simple probability density function. Several of such kernel functions can be composed to form a more advanced density function. Where K satisfies $K(x) \geq 0 \quad \forall x$ and $\int_{\mathbb{R}} K(x)\mathrm{d}x = 1$. One of the commonly used kernel functions is a standard Gauss normal distribution $K(x) \sim N(0, 1)$. The bandwidth h describes how smooth $\hat{f}_h$ becomes. The effects of different bandwidth values are shown in Figure 2.1.

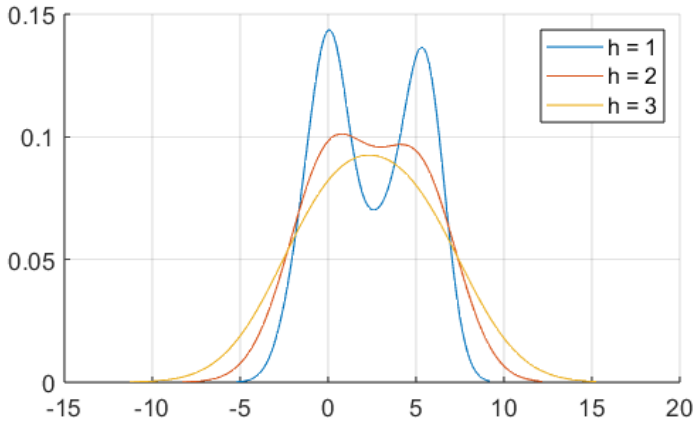


Figure 2.1: Gauss kernel with different bandwidth values.

As stated in [1] there are multiple methods to determine the bandwidth value. One approach is to minimize the expected mean squared integrated error

$$\mathbb{E}\left[\int \left(\hat{f}_h(x) - f(x)\right)^2 \mathrm{d}x\right]. \tag{6}$$

A method to minimize (6) cannot be directly implemented since the distribution of $f(x)$ is unknown. A rule of thumb was proposed by Bernard Silverman [4]. Let $h = 1.06 \min\{\hat{\sigma}, \mathrm{IQR}/1.34\} N^{-1/5}$, where $\hat{\sigma}$ is the sampled standard deviation for x_i and IQR is the difference between the 0.25 quantile and 0.75 quantile in the

sampled set.

2.3 Sensitivity Analysis

Another aspect is to analyze the sensitivity of the model. We want to find how the variance of the model output depends on some specific input parameters. This lets us see how accurate our parameter estimation has to be, to still obtain a reasonable model output. One method for global sensitivity analysis is variance based methods, where Sobol indices are one of the most prominent methods.

2.3.1 Sobol Indices

Sobol indices provides a method for calculating the sensitivity of a model by calculating the variance of the output and then attributing certain amounts of that variance to some set of parameters of the model. Let us summarize the method for Sobol indices outlined in "Global sensitivity indices for nonlinear mathematical models and their Monte Carlo estimates" [3]. The basis of the method is the following ANOVA-decomposition. Let $\mathcal{D} = \{1, \ldots, d\}$, where d is the dimension of the model parameters and a random variable x_i is a parameter $i \in \mathcal{D}$. All x_i are independent and uniformly distributed on the interval $[0, 1]$. Then the function $f(x)$ can be written as

$$f(x) = \sum_{u \subseteq D} f_u(x), \tag{7}$$

where

$$f_u(x) = \int_{[0,1]^{d-|u|}} \left(f(y) - \sum_{v \subset u} f_v(y) \right) \mathrm{d}y. \tag{8}$$

Here y are all the parameters that are not in u ($y = \mathcal{D} \setminus u$). The correlation caused by one certain subset u can then be calculated with

$$\int_{[0,1]^{|u|}} f_u(x)^2 \mathrm{d}x. \tag{9}$$

The computational method proposed in Sobols original paper [3] is to take a subset of parameters whose variance we want to study. The variance of the sum of the combinations of parameters in that subset is calculated. We also calculate the same variance for the remaining parameters. Let y be a subset of the parameters

and z be another subset of the parameters such that $z \cap y = \emptyset$ and $z \cup y = \mathcal{D}$. The variances mentioned above are calculated as

$$D_y = \int f(x) f(y, z') \mathrm{d}x \mathrm{d}z' - f_0^2, \tag{10}$$

$$D_z = \int f(x) f(y', z) \mathrm{d}x \mathrm{d}y' - f_0^2, \tag{11}$$

where

$$f_0 = \int f(x) \mathrm{d}x. \tag{12}$$

The total variance is

$$D = \int f(x)^2 \mathrm{d}x - f_0^2, \tag{13}$$

where the Sobol indices for the set y and z are defined as

$$S_y = D_y / D \in [0, 1] \quad \text{and} \quad S_z = D_z / D \in [0, 1]. \tag{14}$$

These indices are called the principle Sobol indices since they represent the percentage the sets variance contribute to the total

$$D = \sum_{u \subseteq \mathcal{D}} D_u. \tag{15}$$

Approximating integrals with Monte-Carlo integration:

$$\frac{1}{N} \sum^{N} f(x_i) \overset{N \to \infty}{=} \int f(x) \mathrm{d}x. \tag{16}$$

The computational algorithm is as follows:

1. Select a length of samples N and take two independent set of samples of the parameters $A = (y, z)$ and $B = (y', z')$.
2. Use A to calculate D (13) and f_0 (12) using (16).
3. Use A and z' from B to calculate D_y (10) using (16).
4. Use A and y' from B to calculate D_z (11) using (16).

2.4 Inverse Uncertainty Quantification

The goal of inverse uncertainty quantification is to construct a better estimation of the distribution of the model parameters after we have observed some data. There are multiple methods to accomplish this, but we focus on the Maximum Likelihood method in this thesis.

2.4.1 Maximum Likelihood Estimation

Let us assume that the model parameters are distributed in accordance with a multivariate normal distribution. We want to estimate this distribution using the maximum likelihood method outlined in [2]. Assume that γ_{true} are the correct parameters of the model. Let $F_i(\gamma_{\text{true}}, t), i = 1, \ldots, k$ be the simulated output of the model. We consider the measured data

$$\hat{F}_{i,n} = F_i(\gamma_{\text{true}}, t_n) + E_{i,n}, \tag{17}$$

where the error $E_{i,n} \sim N(0, \sigma^2)$ are independent normal random variables. The variance σ^2 has to be estimated based on the measured data. The likelihood function can be described as the probability of observing the errors in the measured data compared to the true data

$$\begin{aligned}\mathcal{L}(\gamma) = &\prod_{n=1}^{N} \frac{1}{2\pi\sigma^2} \exp\left(-\frac{1}{2\sigma^2}(F_1(\gamma, t_n) - \hat{F}_{1,n})^2\right) \times \cdots \times \\ &\times \prod_{n=1}^{N} \frac{1}{2\pi\sigma^2} \exp\left(-\frac{1}{2\sigma^2}(F_k(\gamma, t_n) - \hat{F}_{k,n})^2\right).\end{aligned} \tag{18}$$

We introduce the negative log-likelihood function

$$NLL(\gamma) = -2\log(\mathcal{L}(\gamma)),$$

which becomes

$$NLL(\gamma) = (-2) \cdot \left(\sum_{n=1}^{N} \log(\text{constant terms}) - \frac{(F_1(\gamma, t_n) - \hat{F}_{1,n})^2}{2\sigma^2} + \cdots + \right.$$
$$\left. + \log(\text{constant terms}) - \frac{(F_k(\gamma, t_n) - \hat{F}_{k,n})^2}{2\sigma^2} \right). \tag{19}$$

The maximum likelihood estimator of the model parameters is

$$\gamma_{\text{MLE}} = \text{argmin}(NLL(\gamma)), \tag{20}$$

which can be rewritten as

$$\gamma_{\text{MLE}} = \text{argmin} \left(\sum_{n=1}^{N} \frac{(F_1(\gamma, t_n) - \hat{F}_{1,n})^2}{\sigma^2} + \cdots + \frac{(F_k(\gamma, t_n) - \hat{F}_{k,n})^2}{\sigma^2} \right). \tag{21}$$

This minimization problem is not practical to solve analytically. We apply numerical optimization methods, and in our case we use the Matlab routine `fminsearch` since its a non-linear problem with multi-dimensional parameter and without any strict boundary conditions. Now we have determined an estimate for the mean of the multivariate distribution. Lastly, we also have to determine an estimate for the covariance matrix, which describes the variances and covariances of the model parameters. According to [2] this can be estimated as the inverse of the Fisher Information Matrix, which is calculated as the Hessian of the negative log likelihood function, namely

$$H_{ij} = \frac{\partial^2}{\partial \gamma_i \partial \gamma_j} NLL(\gamma), \tag{22}$$

and the covariance matrix is the inverse

$$\Sigma = H^{-1}. \tag{23}$$

Since the partial derivatives defining H cannot be calculated analytically, we will use a numerical method. We make the following approximation of the Hessian

$$H \approx J^T J, \tag{24}$$

where J is the gradient of of $NLL(\gamma)$ and the derivatives are in turn calculated using a standard finite difference quotient approximation. As an example, let $f : \mathbb{R}^2 \to \mathbb{R}$ and select a discretization step Δx. Then the finite difference qoutient is

$$\frac{\partial f(x,y)}{\partial x} \simeq \frac{f(x+\Delta x, y) - f(x,y)}{\Delta x}.$$

3 Method

We want to compare our two models, (1) and (2). This will be done by estimating some statistical properties of the quantities of interests. We follow the method outlined in [2] where we firstly consider our model (SIR and SEIR) with a certain prior interval of their parameters.

Secondly, we want to determine which of the parameters are the most important for the quantities of interests. This is done using Sobol indices, where we partition the parameters γ of the models into the following two sets $\{\gamma_i\}$ and $\{\gamma_j, \forall j \neq i\}$. We construct all such sets and determine which parameters are the most significant.

Synthetic data is used in the inverse uncertainty quantification, instead of real world data. This allows to understand the error sources more clearly, leading to a clearer analysis. The synthetic data is generated by adding noise to the simulated trajectories, given some chosen model parameters. The added noise is normally distributed with a mean of 0 and standard deviation σ.

We apply the maximum likelihood method to estimate the distribution of the model parameters. In the case of the SIR-model we only observe the infected and recovered compartments, as these are the values that are usually obtainable. Additionally, for the SEIR-model we observe the exposed compartment.

We apply the forward uncertainty quantification on this updated parameter distribution and obtain probability density functions for the quantities of interests.

In the results section we make use of the following conditions. From [2] the parameters used for SIR model is in the interval $\beta \in [0.25, 0.35]$ and $r \in [0.06, 0.18]$. Since the distribution of these variables are unknown the used priori distribution is uniform. The additional parameter in the SEIR model is i, which is uniformly distributed on $[0.14, 0.33]$. In order to simplify the model representation the total population $N_{pop} = 1$ in (3) and (4). This gives values for S, E, I and R to be between 0 and 1, which is interpreted as the proportion of the population instead.

4 Results

The numerical results are based on the initial conditions listed in table 4.1

	S_0	E_0	I_0	R_0
SIR-model	0.95		0.05	0
SEIR-model	0.95	0.02	0.03	0

Figure 4.1: Initial conditions used in the simulations

When generating synthetic data, we use $\beta = 0.30$, $r = 0.12$ and $i = 0.24$ as the true parameters of the models. For both the models we consider the peak time for infections and the peak value for infections as quantities of interests. Additionally, we study the peak time for exposed and the peak value for exposed in the SEIR-model (2).

4.1 SIR

Figure 4.2 shows the trajectories for the components S, I and R with the parameters being uniformly distributed based on the Monte Carlo method with $N = 250$. The black solid lines are the mean value of each component, while the black dashed line represent the simulation for the SIR-model with the true parameters.

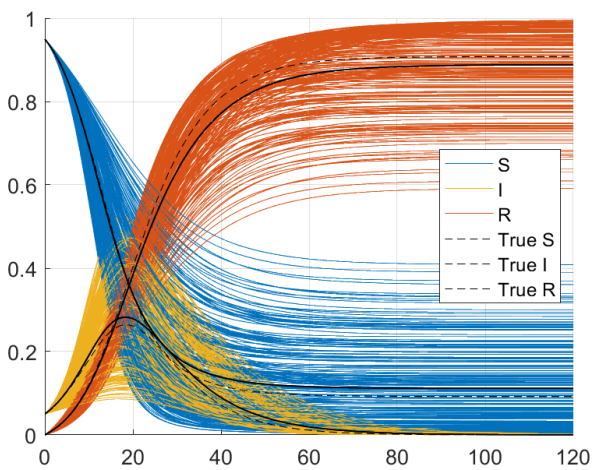


Figure 4.2: SIR-model Monte-Carlo sampling with $N = 250$ on priori PDF.

Figure 4.3 presents the principal Sobol indices for the infected peak time and the peak value. We compute the Sobol indices using $N = 1\,000\,000$ samples.

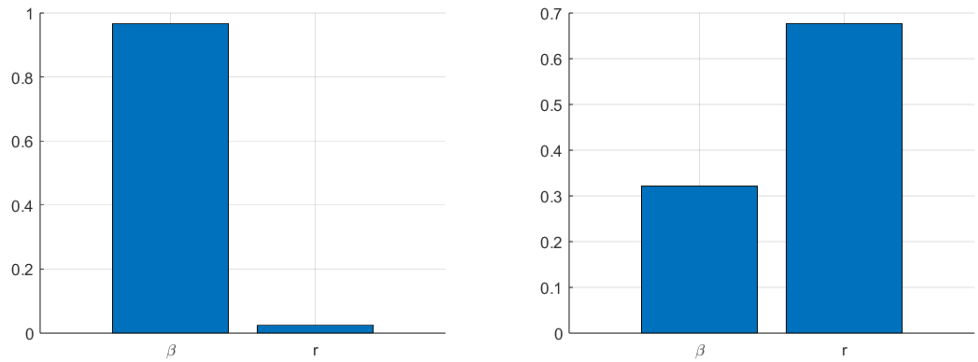


Figure 4.3: Principle Sobol indices for Infected peak time (left) and peak value (right) for the SIR-model.

Let us analyze the distribution of the infected peak time and peak value in Figure 4.2. We compute a kernel density estimation using the standard normal kernel $K \sim N(0, 1)$ and Silverman's rule for bandwidth selection. We obtain bandwidths 0.0411 and 0.520 respectively. Figure 4.4 presents the obtained PDF:s, and the dotted vertical line displays the value of the quantity of interest given the true parameter values.

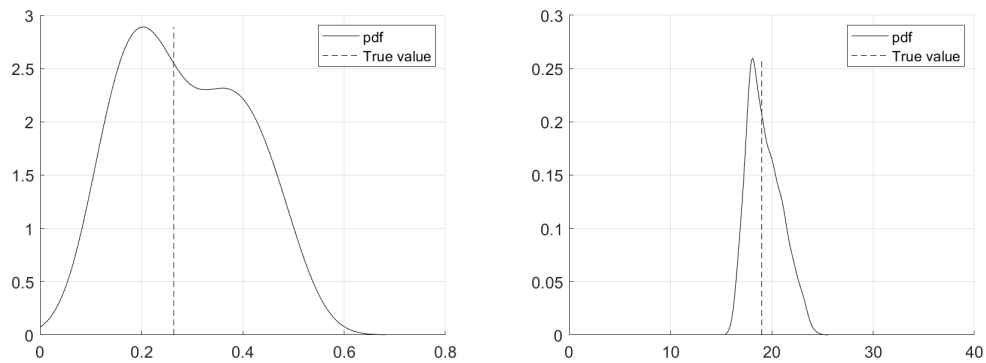


Figure 4.4: Kernel density functions for the peak value (left) and peak time (right).

We create the synthetic data used in the inverse uncertainty quantification by running the simulation with the true values $\beta = 0.30$, $r = 0.12$ and adding Gaussian noise with a standard deviation of $\sigma = 0.05$. We calculate a maximum likelihood approximation $\widehat{\mu}$ of the model parameters. Figure 4.5 plots the results.

The solid line is the SIR-model simulated using $\widehat{\mu}$ and the dashed line is the simulation using the underlying true parameters. The synthetic data is plotted as a circled line. The following estimation is computed

$$\widehat{\mu} = [\beta\, r] = [0.305\, 0.119], \quad \Sigma = 10^{-5} \begin{bmatrix} 3.55 & 1.32 \\ 1.32 & 2.50 \end{bmatrix}. \tag{25}$$

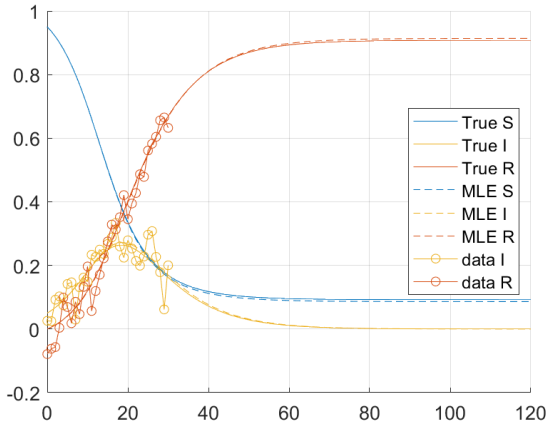


Figure 4.5: Synthetic data plotted together with the true value and MLE simulations.

Figure 4.6 plots the multivariate normal distribution for β and r that we obtain using the maximum likelihood estimation (25).

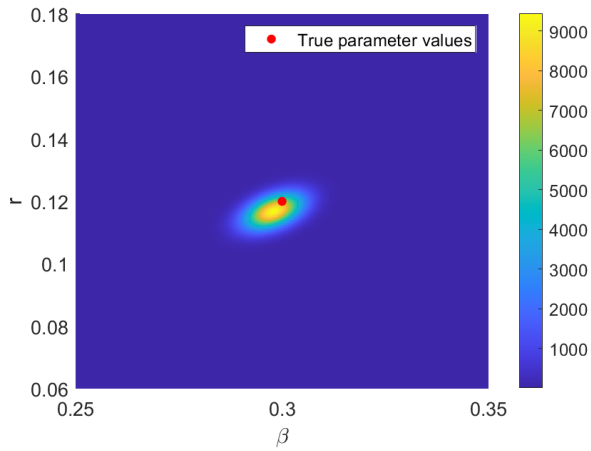


Figure 4.6: Multivariate normal distribution for β and r with parameters from (25).

In Figure 4.7 we sample the parameters from the multivariate normal distribution (25) and simulate the trajectories of the SIR-model.

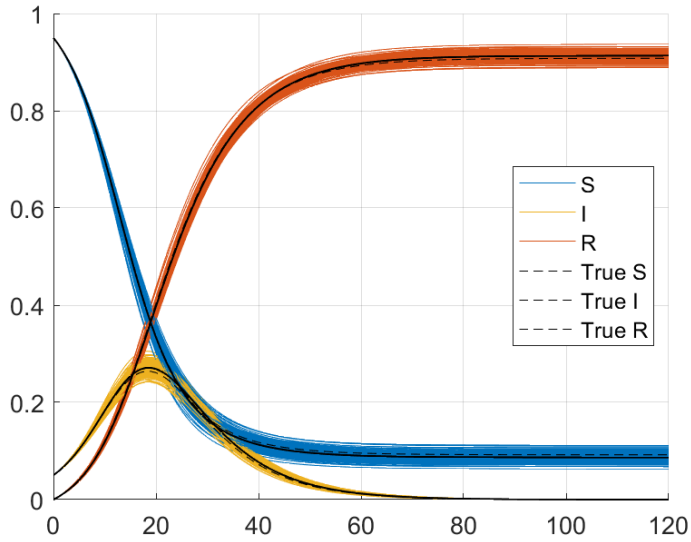


Figure 4.7: SIR-model Monte-Carlo sampling with N = 250 on posterior PDF with parameter values (25).

We want to estimate the probability density function of the infected peak time and peak value. We apply the same method as in Figure 4.4. We acquire the bandwidths 0.00442 and 0.702. Figure 4.8 plots the PDF:s we obtain using the kernel density estimation.

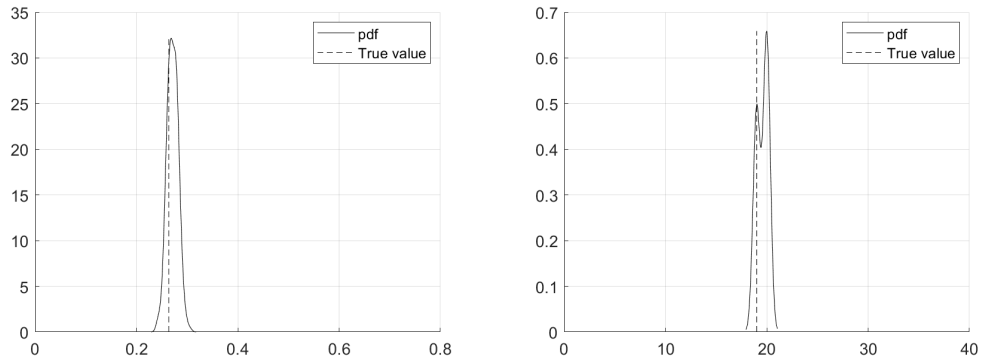


Figure 4.8: Updated Kernel density functions for the peak value (left) and peak time (right).

4.2 SEIR

Figure 4.9 shows the trajectories for the components S, E, I and R with the parameters being uniformly distributed based on the Monte Carlo method with $N = 250$. The black solid lines represent the mean value of each component, and the black dashed lines represents the simulation for the SEIR-model with the true parameters.

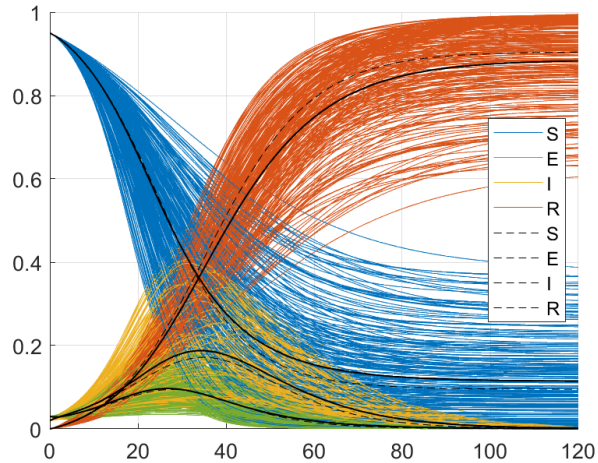


Figure 4.9: SEIR-model Monte-Carlo sampling with N = 250 on priori PDF.

Figure 4.10 presents the principal Sobol indices for the infected peak time and peak value. Figure 4.11 presents the principal Sobol indices for the exposed peak time and peak value. We compute all of these Sobol indices using $N = 1\,000\,000$ samples.

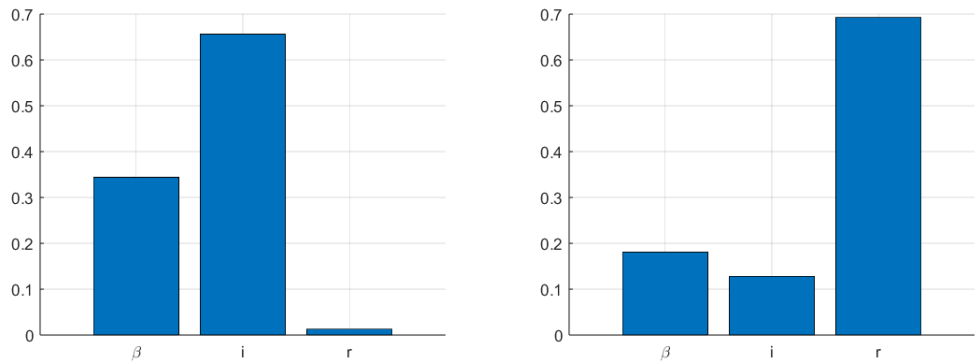


Figure 4.10: Principle Sobol indices for infected peak time (left) and peak value (right) for the SEIR-model.

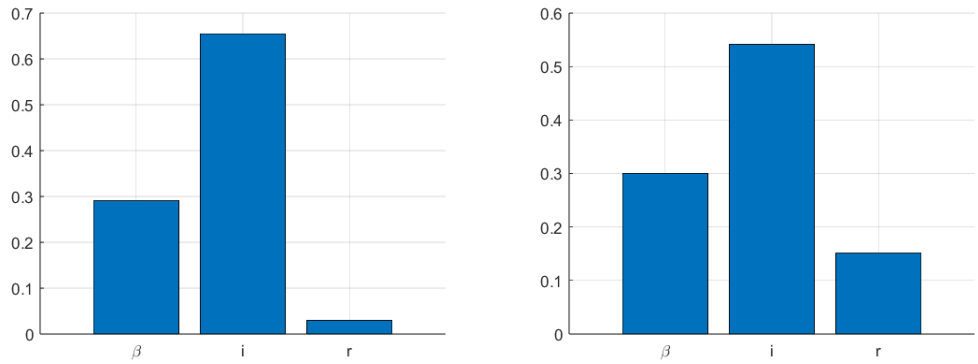


Figure 4.11: Principle Sobol indices for exposed peak time (left) and peak value (right) for the SEIR-model.

Next, we analyze the distribution of the quantities of interests. As for the SIR-model, we compute a kernel density estimation using a standard normal kernel and Silverman's rule for bandwidth selection. Figure 4.12 shows the estimated probability density functions for infected peak time and value. The vertical line shows the calculated value of the quantity of interest using the true parameters. Figure 4.13 shows the estimated probability density functions for the exposed peak time and value. The vertical line has a similar meaning as for Figure 4.12.

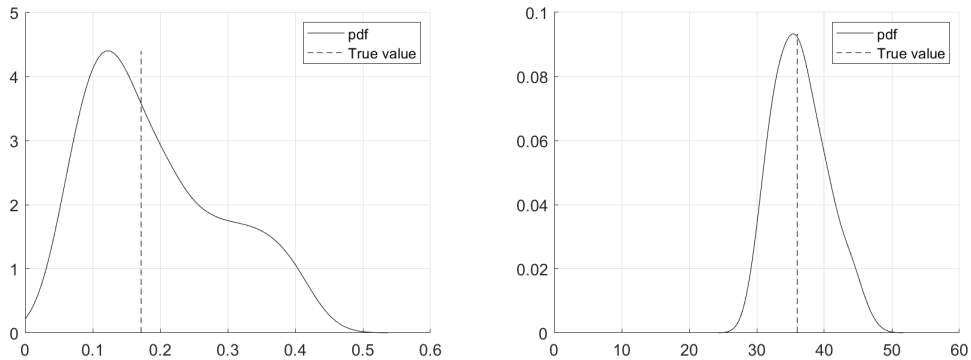


Figure 4.12: Kernel density functions for infected peak value (left) and peak time (right). Using the bandwidths 0.0342 and 1.56.

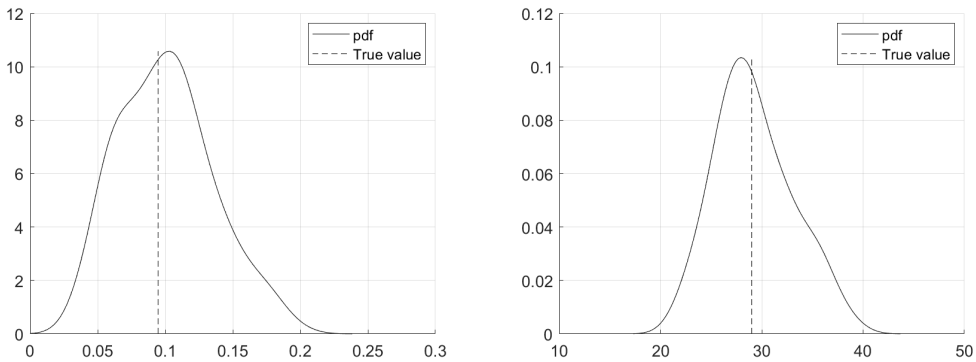


Figure 4.13: Kernel density functions for Exposed peak value (left) and peak time (right). Using the bandwidths 0.0130 and 1.56.

We create synthetic data by generating a reference solution for the trajectories using the true parameters mentioned above. Gaussian noise is sampled with a standard deviation of $\sigma = 0.025$ and added to the reference solution. The following maximum likelihood estimation is obtained

$$\hat{\mu} = \begin{bmatrix} \beta \\ r \\ i \end{bmatrix} = \begin{bmatrix} 0.301 \\ 0.124 \\ 0.248 \end{bmatrix}, \quad \Sigma = 10^{-4} \begin{bmatrix} 0.974 & 0.344 & -0.989 \\ 0.344 & 0.236 & -0.216 \\ -0.989 & -0.216 & 1.80 \end{bmatrix}. \tag{26}$$

We simulate the trajectories using the parameters $\widehat{\mu}$ obtained from the maximum likelihood estimation. Figure 4.14 plots the reference solution with a solid line. The trajectories obtained with the MLE parameters are plotted with a dashed line. The synthetic data is plotted with a circled line.

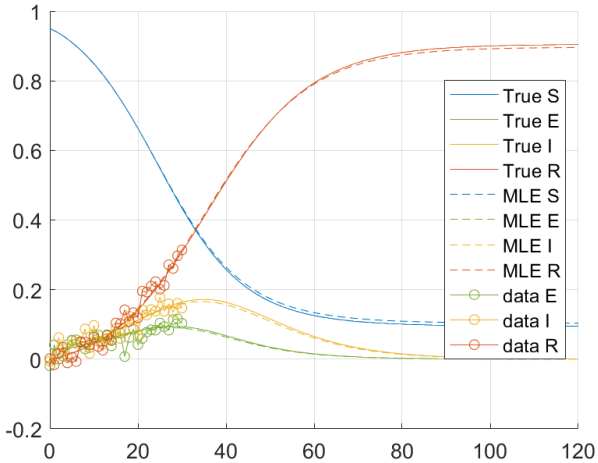
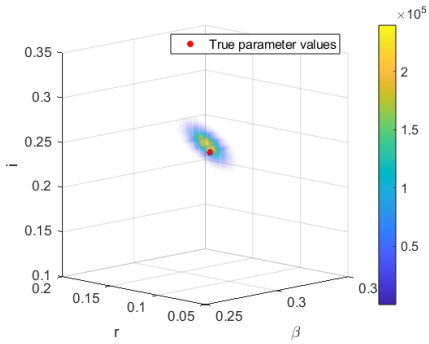
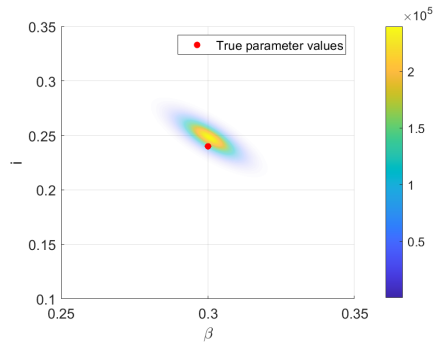


Figure 4.14: Synthetic data plotted together with the true value and MLE simulations.

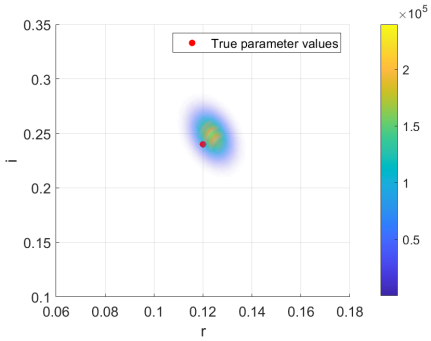
Figure 4.15 plots the multivariate normal distribution that we obtain for the parameters using the maximum likelihood estimation in a slice plot.



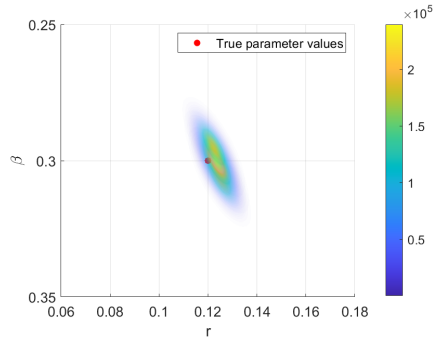
(a) 3D view of the multivariate normal distribution.



(b) Side view for the cross section between β and i.



(c) Side view for the cross section between r and i.



(d) Side view for the cross section between β and r.

Figure 4.15: Multivariate normal distribution with parameters being the MLE estimation from (26).

In Figure 4.16 we sample the parameters from the distribution in Figure 4.15. Then we plot the trajectories of the components.

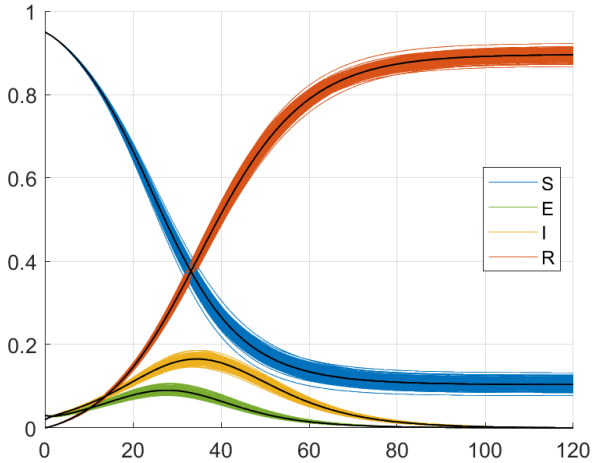


Figure 4.16: SEIR-model Monte-Carlo sampling with N = 250 on posterior PDF with parameter values (26)

.

We estimate the probability density function of the infected peak time and peak value. We apply the same method as in Figure 4.4. We calculate the bandwidths 0.00248 (left) and 0.520 (right). Figure 4.17 plots the PDF's we obtain using the kernel density estimation.

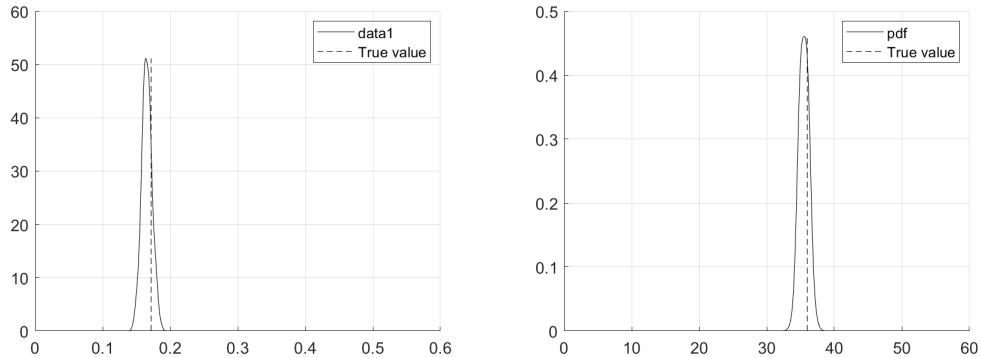


Figure 4.17: Updated Kernel density functions for infected peak value (left) and peak time (right).

Additionally, we estimate the probability density function of the exposed peak time and peak value. The bandwidths are 0.00194 (left) and 0.702 (right). Figure 4.18 plots the PDF's we obtain using the kernel density estimation.

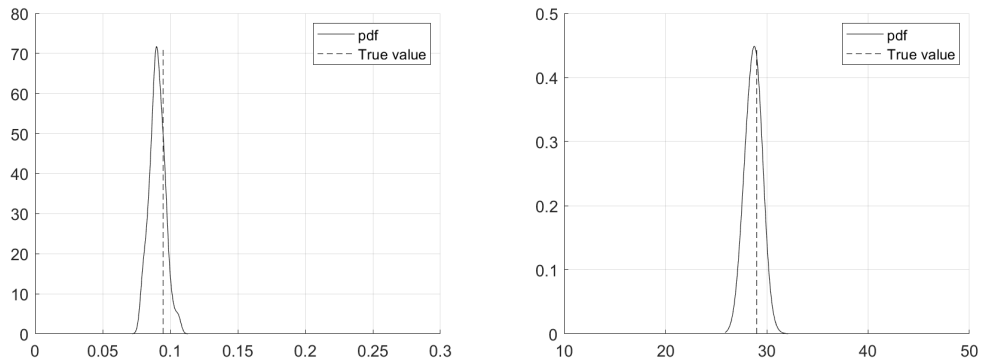


Figure 4.18: Updated Kernel density functions for Exposed peak value (left) and peak time (right).

4.3 SEIR data in SIR model

We generate a reference solution using the SEIR-model with parameters $\beta = 0.3$, $r = 0.12$ and $i = 0.24$. A Gaussian noise with a standard deviation of $\sigma = 0.05$ is added. The parameters of the SIR-model is estimated using the maximum likelihood method. Figure 4.19 plots the true trajectories with solid lines, and the estimated trajectories with dashed lines. Additionally, the synthetic data points are plotted using the circled lines.

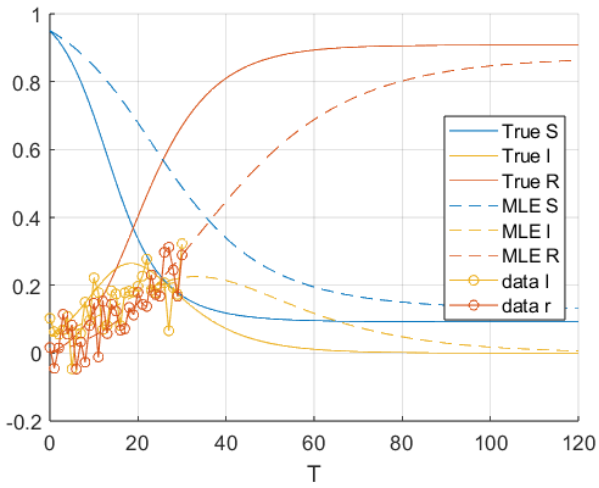


Figure 4.19: Synthetic data plotted together with the true value and MLE simulations.

5 Conclusion

With the prior uniform distribution there is a large variation in the trajectories of the SIR- and SEIR-model, as can be seen in Figure 4.2 and Figure 4.9, which leads to the large probability spread for the quantities of interest, peak value and peak time for infected and exposed seen in Figure 4.4, Figure 4.12, and Figure 4.13. Without any knowledge of the parameter distributions or their relation to each other it is hard to draw any conclusions from these density functions as to the nature of the quantities of interest.

We have computed the Sobol indices for the peak time and value of infections for both the SIR- and SEIR-model. We notice that the r parameter has a very small impact on the output of the peak time of both the models. Whereas for the peak value of the models r is instead the most important parameter. In other words, the choice of quantity of interest has a large impact on which model parameters are the most important.

After estimating the model parameters based on a set of synthetic data using maximum likelihood estimation, the time evolution of the SIR system is plotted together with the synthetic data in Figure 4.5 and Figure 4.14. It yields the multivariate normal distribution shown in Figure 4.6 with the parameters (25) and Figure 4.15 with parameters (26). We plot the trajectories for these distributions in Figure 4.7 for the SIR model and Figure 4.16 for the SEIR model. Here we see that the trajectories are much more constrained around the reference paths. The kernel density estimates of the density functions seen in Figure 4.8 are also much more narrow, indicating a decreased variance in the quantities of interest. How precisely the maximum likelihood approximation fits the measured data (synthetic or observed) depends on how much the actual disease fits the compartmental model. Differences in the actual distribution of the measurement error is another possible source of error. We have assumed a normally distributed measurement error, which might not be the case.

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